Sprint 5 Report

Youtube Link: https://youtu.be/BfjVvXwHhAE

GitHub Link: https://github.com/KieranMcHugh/WSUDAS-AI.git

What's New (User Facing)

After the introduction of the new model in the previous sprint, this new cycle is about integrating the model into the existing system, providing a seamless product as a whole. We have begun the process of implementing the system into WSUDAS' service, including the setup for pushing a csv file and returning the data as a JSON file

Work Summary (Developer Facing)

Building from the momentum of last sprint, this sprint is about creating API endpoints for the model, converting the R part into python for a uniform language project, and easier to run in a FastAPI environment. Last but not least is to integrate the API into the current system as a Docker container. This would allow it to run seamlessly with the rest of WSUDAS' services and provide the location for where the system itself will exist.

Unfinished Work

We have made progress in all parts of the project but none of them are fully completed as of now. There is one API endpoint for running the model, but the integration as a Docker container is not fully done yet. The feature extraction in R is mostly done, though there is still some debugging required to resolve a minor problem. The feature extraction output is off by very small decimal amounts, which appears to translate to a widely different range in terms of the LT50's output. We are currently investigating why this is the case and will then correct the feature extraction to amend this.

Completed Issues/User Stories

Here are links to the issues that we completed in this sprint:

* Depickling Python Files: https://github.com/users/KieranMcHugh/projects/3/views/1?pane=issue&itemId=157956505&issue=KieranMcHugh%7CWSUDAS-AI%7C14
* Reformat extraction code to accept WSUDAS' data: https://github.com/users/KieranMcHugh/projects/3/views/1?pane=issue&itemId=157956201&issue=KieranMcHugh%7CWSUDAS-AI%7C11

Code Files for Review

Please review the following code files, which were actively developed during this sprint, for quality:

*[feature_extraction.py]
https://github.com/KieranMcHugh/WSUDAS-AI/blob/main/sample_code/feature_extraction.py
*[WEL2_extract.ipynb]
https://github.com/KieranMcHugh/WSUDAS-AI/blob/main/sample_code/WEL2_extract.ipynb
*[sample.ipynb]
https://github.com/KieranMcHugh/WSUDAS-AI/blob/main/sample_code/sample.ipynb

Retrospective Summary
Here's what went well:
* Item 1: We now have an API endpoint for the model
* Item 2: We successfully created a container for the API endpoint to be up and running inside Docker
* Item 3: We successfully rewrote the feature extraction in Python from R with very minimal differences from the original.

Here's what we'd like to improve:
* Item 1: There are still some problems for getting the container to run seamlessly everytime docker is started.
* Item 2: The feature extracted from Python is off by a little compared to R, which translates to a different result of the LT50.

Here are changes we plan to implement in the next sprint:
* Fully address any remaining goals from this sprint (i.e. resolving feature extraction discrepancy and setting up docker completely).
* Start the training process on our internal data instead of using their sample data